PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 1 • Chapter OL-T20 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

2 classified items • 7 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T20 • Expertise Q16+ • 2 items • 7 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Completing the Square

Unit 1 • Expertise • 7 marks • 2 item(s)

June 2023 | 4MA1-2H Q19 | 3 marks

Positive non-unit leading coefficient

January 2022 | 4MA1-1H Q20 | 4 marks

Negative leading coefficient or maximum form

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q19

3 marks

19 Express $3x^2 - 6x + 5$ in the form $a(x-b)^2 + c$

....

WORKING SPACE

4MA1-2H Q19 **3 marks**

19 Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

....

Complete the square **3 marks**

Group the quadratic terms

$$3x^2 - 6x + 5 = 3(x^2 - 2x) + 5.$$

Complete the square

$$3(x^2 - 2x) + 5 = 3[(x - 1)^2 - 1] + 5 = 3(x - 1)^2 + 2.$$

Final answer

$$3(x - 1)^2 + 2$$

4MA1-1H Q20 Q20(a), Q20(b) **4 marks**

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a, b and c are integers.

....
(3)

C is the curve with equation $y = 7 + 12x - 3x^2$
The point A is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of A

(...................... , ...)
(1)

WORKING SPACE

4MA1-1H Q20 Q20(a), Q20(b) **4 marks**

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(1)

Chapter: Completing the Square **4 marks**

Q20(a) – Chapter: Completing the Square

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

Method
$7+12x-3x^2=7-3(x^2-4x)$.
Why: Factor −3 from the quadratic terms.

Method
$x^2-4x=(x-2)$^2−4.
Why: Complete the square.

Method
Therefore $7+12x-3x^2=19-3(x-2)$^2.
Why: Substitute the completed square and simplify.

Final answer
completed square: 19−3(x−2)^2

Q20(b) – Chapter: Completing the Square

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

Method
The maximum occurs when (x−2)^2=0, giving A=(2,19).
Why: The negative coefficient means the squared term is subtracted from 19.

Final answer
maximum point: (2,19)